

WORKPACK 2

Is This Data Honest?

Level 3 Data Technician · GreenWheels Urban Mobility · Self-Diagnostic Pack

Get Ready to Smash This

A head-start check — this is not a test

Below are eight things this pack covers. Tick the ones you can already do. Each tick shows how much of a head start you have.

If you can tick all eight: skip to the Mini-Mission on page 8 and prove it in two minutes. Then move on to Pack 3.

- ☐ I can sort a column from smallest to largest, and from largest to smallest.
- ☐ I can use AutoFilter to show only rows that meet a condition I choose.
- ☐ I can filter a column to show only rows where the value is zero.
- ☐ I know the difference between a blank cell and a cell that contains zero.
- ☐ I can use the data dictionary to check whether a value is inside the valid range.
- ☐ I can sort two columns at the same time to check for inconsistencies.
- ☐ I know what an anomaly is and why you document it before you change anything.
- ☐ I can record a finding in an anomaly log with a Record ID, column name, and description.
- ☐ I can find duplicate records using Conditional Formatting or a COUNTIFS formula.
- ☐ I can find blank cells using Go To Special or a column filter.

Marcus from the next desk says: "I validated the whole dataset. I sorted by Record ID and scrolled through it — everything looks fine to me." (We will come back to this.)

What This Pack Is About

By the end of this pack, you will be able to sort and filter the GreenWheels dataset to find anomalies — values that are missing, impossible, or inconsistent. You will also be able to document each finding in an anomaly log. This is Deliverable 1 in the Scenario Demonstration.

The pack takes around 30 to 40 minutes if you work through every section. You can pause after any skill chunk and come back later.

The Eight Anomalies in This Dataset

The GreenWheels dataset contains eight problems. You will find them one by one during this pack. Here is a summary of what you are looking for — but do not skip ahead. The point is to practise finding them yourself.

Record ID	Station	Column	Issue Found	Action
HC-010	Riverside Walk	Total Revenue (£)	£0 revenue recorded — 487 hires in same quarter	Flag: investigate
HC-015	Market Street	Total Hires	1,847 hires — data dictionary valid range is 100–800	Flag: investigate
HC-019	University Gate	Active Bikes	0 bikes but 521 hires and £4,429 revenue recorded	Flag: contradiction
HC-030	South Station	Zone	Shows 'West' — all other South Station records show 'South'	Flag: entry error
HC-031	South Station	Total Hires	8 hires — all other quarters at this station: 389–478	Flag: investigate
HC-035	Canal Quarter	Avg Hire Duration (Mins)	0 minutes — data dictionary states 0 is invalid	Flag: invalid value
HC-041	Victoria Park	Station + Period	Duplicate: same station and quarter as HC-007 (Q3 2024)	Flag: remove duplicate
HC-042	Meadow Park	Total Hires / Revenue / Duration / Satisfaction	Four blank numeric fields — data not captured for this station	Flag: missing values

This table is also your model for Deliverable 1. In the demonstration, you will produce a log like this for the full dataset. Two of the eight anomalies — the duplicate and the blank record — are covered in Chunks 6 and 7.

The Skills

Each chunk covers one validation technique. The same real dataset is used throughout, so every anomaly you find is a real issue in the actual GreenWheels data.

1

Sorting to Spot Outliers

Priya wants a quick sense-check

SCENARIO

Priya has asked you to do a quick sense-check on Total Hires before the analysis begins. "Anything that looks wildly wrong, flag it. I just need to know the data is roughly right."

Sorting a column puts its values in order — smallest to largest, or largest to smallest. This brings the most extreme values to the top or bottom of the list, where they are easy to spot.

WORD TO KNOW

Sort

To arrange all rows in a dataset in order based on one column's values. Sorting does not delete or change any data — it only changes the order in which rows are displayed.

WORD TO KNOW

Ascending / Descending

Ascending order goes from smallest to largest (1, 2, 3 — or A, B, C). Descending order goes from largest to smallest (100, 50, 10 — or Z, A). In Excel, the sort buttons look like A→Z and Z→A.

To sort by a column: click any cell in that column, then go to the Data tab on the ribbon and click the A→Z button (ascending) or Z→A button (descending).

TRY IT

Open `urban_cycle_hire_aggregated.xlsx`. Click any cell in the Total Hires column. Sort ascending (A→Z). Look at the rows at the top — which stations have the lowest hire counts?

Now sort descending (Z→A). Look at the row at the very top. Check the Total Hires value against the valid range in the Data Dictionary (100–800 per quarter).

ANOMALY FOUND

Ascending sort reveals: HC-031 (South Station, Q3 2024) — only 8 hires. HC-034 (Canal Quarter, Q2 2024) — only 45 hires. Both are far below the typical range.

Descending sort reveals: HC-015 (Market Street, Q3 2024) — 1,847 hires. The Data Dictionary states the valid range is 100–800. This record is more than double the maximum.

CHECK YOURSELF

After sorting Total Hires descending, which record sits at the top, and why is its value a problem?

Answer: HC-015 (Market Street, Q3 2024) with 1,847 hires. The Data Dictionary states the valid range is 100–800 per quarter. 1,847 is more than double the stated maximum and should be flagged for investigation.

2 Filtering for Zeros

Revenue should not be £0

SCENARIO

The Data Dictionary says: "Total Revenue of £0 requires investigation." You want to find any record where revenue is exactly zero.

Scrolling through 40 rows to look for zeroes would work — but what if there were 4,000 rows?

AutoFilter lets you show only the rows that meet a condition you choose. All other rows are hidden temporarily — they are not deleted. A filter arrow appears at the top of each column when AutoFilter is turned on.

WORD TO KNOW

AutoFilter

A tool that hides rows that do not meet your chosen condition, leaving only matching rows visible. Turn it on by clicking anywhere in the data and pressing Ctrl+Shift+L, or by going to Data → Filter.

WORD TO KNOW

Filter arrow

The small dropdown arrow that appears in each column header when AutoFilter is on. Click it to set a condition for that column — for example, show only rows where the value equals 0.

To filter for zero: click the filter arrow on the Total Revenue column → Number Filters → Equals → type 0 → click OK.

TRY IT

Turn on AutoFilter (Ctrl+Shift+L). Click the filter arrow on the Total Revenue (£) column. Choose Number Filters → Equals, type 0, and click OK. How many rows are shown?

Now clear that filter. Apply a new filter on the Avg Hire Duration (Mins) column — again filtering for rows equal to 0. How many rows are shown, and what record is it?

ANOMALY FOUND

Revenue = £0: HC-010 (Riverside Walk, Q2 2024). This record shows 487 hires but £0 revenue — that is impossible if any hires actually took place.

Avg Duration = 0: HC-035 (Canal Quarter, Q3 2024). The Data Dictionary states that 0 is an invalid value for this column.

CHECK YOURSELF

How is filtering for zeros different from just scrolling through the data looking for them?

Answer: Filtering is reliable and repeatable. Scrolling depends on you spotting every issue by eye — easy to miss one, especially across many rows or columns. A filter finds every matching row in one step.

3

Spotting Contradictions

When two values cannot both be right

SCENARIO

The Data Dictionary says Active Bikes = 0 means the station was inactive during that quarter. But what if a record shows Active Bikes = 0 and also shows hundreds of hires and thousands in revenue?

A contradiction is when two values in the same row cannot both be correct. Neither value alone looks wrong — but together they are impossible. Sorting and filtering can reveal contradictions that no single-column check would catch.

WORD TO KNOW

Contradiction

Two values in the same record that cannot both be true at the same time. For example: Active Bikes = 0 (station inactive) alongside Total Hires = 521 (hires took place). Both cannot be correct.

To check for this: filter the Active Bikes column for rows equal to 0. Then look at the Total Hires and Total Revenue columns in those rows.

TRY IT

Clear any current filters. Apply a new filter on the Active Bikes column showing only rows equal to 0. Look at the Total Hires and Total Revenue values in the rows that appear.

ANOMALY FOUND

HC-019 (University Gate, Q3 2024): Active Bikes = 0, but Total Hires = 521 and Total Revenue = £4,429.

A station with no active bikes cannot record hires or revenue. This is a contradiction — at least one of the three values must be wrong.

CHECK YOURSELF

Why would filtering for zeros in the Active Bikes column alone not be enough to spot this anomaly?

Answer: A zero in Active Bikes is only a problem when hires or revenue are also present. Without checking the other columns in the same row, you would not know whether the zero is valid (station genuinely inactive, zero hires) or contradictory (station inactive but hires recorded).

4**Checking Consistency***One station, four records — do they agree?***SCENARIO**

Each station appears four times in the dataset — once per quarter. Values like Station Name and Zone should be identical in all four rows for the same station.

A data entry error in one row can make a station appear to belong to the wrong part of the city.

To check for consistency: sort by Station Name (A→Z). All four rows for each station will appear together. Then scan the Zone column — if one row shows a different zone from the other three, you have found an inconsistency.

WORD TO KNOW**Consistency check**

Verifying that values that should be the same across multiple records actually are the same. A station that appears as 'South' in three rows and 'West' in one row has a consistency problem.

The five valid zone values are: Central, North, South, East, West. Each station belongs to exactly one zone for the whole year.

TRY IT

Clear any current filters. Sort the data by Station Name (A→Z). Scroll through the rows, keeping your eye on the Zone column. Look for any station where one quarter shows a different zone from the other three.

ANOMALY FOUND

South Station: HC-029 (Q1), HC-031 (Q3), and HC-032 (Q4) all show Zone = 'South'. But HC-030 (Q2 2024) shows Zone = 'West'.

South Station is in the South zone. The Q2 2024 record contains a data entry error.

CHECK YOURSELF

How does sorting by Station Name help you find zone inconsistencies?

Answer: Sorting groups all four rows for each station together. This makes it easy to compare the Zone value across all four quarters side by side. Without sorting, the four rows for South Station are spread across the sheet and the inconsistency is much harder to see.

5**Documenting What You Found***Finding it is half the job***SCENARIO**

You have found six anomalies. Before you change or report anything, you need to write down what you found, where it was, and what you think the problem is.

This record becomes Deliverable 1 in the demonstration — and it protects you if anyone later asks why you changed something.

An anomaly log is a table that records each problem you find. It shows the Record ID, the column that contains the problem, a plain-language description of the issue, and what action you took or recommend. Your log for this pack will have eight rows — one for each anomaly.

WORD TO KNOW**Anomaly log**

A table that records every data quality issue you find. Each row describes one problem: where it is, what it is, and what was done about it. The log is evidence that you checked the data carefully.

WORD TO KNOW**Flag**

To mark a record as needing attention, without changing the underlying data. You might flag a record by adding a note, a colour, or a new column called 'Issue' with a short description.

In your demonstration file, create a new sheet called Anomaly_Log. Add four column headers: Record ID, Column, Issue Found, Action. Then add one row for each of the six anomalies you found in this pack.

TRY IT

In the GreenWheels file, add a new sheet and name it Anomaly_Log. Create the four column headers. Add all eight anomalies from this pack — including the duplicate (Chunk 6) and the blank record (Chunk 7). Use the table at the start of this pack as your reference.

CHECK YOURSELF

Why is it important to document anomalies in a log rather than just correcting them straight away?

Answer: Documentation shows what was changed and why. Without it, you cannot explain your decisions to colleagues, managers, or assessors. You also cannot reverse a change if it turns out you made the wrong call.

6**Finding Duplicate Records***The same quarter, twice***SCENARIO***Priya has told you that data was recently merged from two separate booking systems.**"There may be records that got submitted twice — same station, same quarter, slightly different numbers. Can you check?"*

A duplicate record is a row that represents the same real-world event as another row. In this dataset, each station should appear exactly once per quarter. If a station appears twice for the same quarter, one of those rows should not be there.

WORD TO KNOW**Duplicate**

A record that represents the same event as another record already in the dataset. Duplicates inflate totals and distort averages — counting the same quarter's hires twice makes a station look busier than it was.

WORD TO KNOW**Conditional Formatting**

A feature that automatically changes the colour or style of a cell when its value meets a condition you set. It is useful for making duplicates visible at a glance without changing any data.

Method 1 — Highlight duplicate values: Select the Hire Record ID column. Go to Home → Conditional Formatting → Highlight Cells Rules → Duplicate Values. Any Record ID that appears more than once will be highlighted.

Method 2 — Count combinations: Add a temporary helper column next to the data. In the first data row, type the formula `=COUNTIFS($B$2:$B$43,B2,$D$2:$D$43,D2)`. This counts how many rows share the same Station Name and Period. A result greater than 1 means that combination appears more than once.

WORD TO KNOW**COUNTIFS**

A formula that counts rows where multiple conditions are all true at the same time. `=COUNTIFS(range1, value1, range2, value2)` counts rows where range1 matches value1 AND range2 matches value2.

TRY IT

Select the Hire Record ID column (column A, rows 2 to 43). Apply Conditional Formatting → Highlight Cells Rules → Duplicate Values. Look for any highlighted cells.

Now try Method 2. In an empty column next to row 2, type:

`=COUNTIFS($B$2:$B$43,B2,$D$2:$D$43,D2)` and press Enter. Copy the formula down to row 43. Look for any row where the result is 2 or more.

ANOMALY FOUND

HC-041 (Victoria Park, Q3 2024) is a duplicate of HC-007. Both records share the same Station Name and Period. The values are slightly different — 445 vs 451 hires — because the data came from two different systems.

CHECK YOURSELF

Why is it not enough to look for duplicate Record IDs only — why must you also check Station Name and Period?

Answer: HC-041 has a unique Record ID, so it would not be flagged by a simple ID check. The duplication is in the combination of Station Name and Period — two rows claiming to be Q3 2024 data for Victoria Park. The ID check misses this entirely.

7**Finding Blank Cells***When the data never arrived***SCENARIO**

Meadow Park is a new station that joined the GreenWheels network in Q4 2024. Its hire tracking system was not ready in time.

A record was submitted with the station details and bike allocation — but the operational data fields were left blank.

A blank cell in a numeric column is different from a zero. Zero means the value was measured and the result was nothing. Blank means the value was never captured at all. A formula that calculates an average will treat blanks differently from zeros — and both can give misleading results if not handled.

WORD TO KNOW**Blank cell**

A cell that contains no value at all. In calculations, Excel often treats blanks as zero — but not always. AVERAGE ignores blanks, which can make averages look higher than they should be. COUNT ignores blanks too. COUNTA counts them because it counts non-empty cells.

WORD TO KNOW**Go To Special**

An Excel feature that selects all cells of a specific type at once. Go to Home → Find & Select → Go To Special, then choose Blanks. Excel selects every blank cell in your current selection, making them easy to see and count.

Method 1 — Go To Special: Select the whole dataset. Go to Home → Find & Select → Go To Special → Blanks. All blank cells are selected and highlighted. Count them in the status bar at the bottom of the screen.

Method 2 — Filter for blanks: Click the filter arrow on any numeric column. In the filter list, scroll to the bottom and tick (Blanks). Rows with blank values in that column appear.

Method 3 — COUNTBLANK formula: In an empty cell, type =COUNTBLANK(F2:F43) to count blanks in the Total Hires column. Repeat for each numeric column you want to check.

WORD TO KNOW**COUNTBLANK**

A formula that counts the number of blank cells in a range. =COUNTBLANK(F2:F43) returns the number of empty cells in the Total Hires column.

TRY IT

Select all the data (A1:J43). Go to Home → Find & Select → Go To Special → Blanks. Note how many cells are selected and which rows they are in.

Now try Method 2. Turn on AutoFilter (Ctrl+Shift+L). Click the filter arrow on the Total Hires column. Scroll to the bottom of the filter list and tick (Blanks). Which record appears?

ANOMALY FOUND

HC-042 (Meadow Park, Q4 2024) has four blank numeric fields: Total Hires, Avg Hire Duration, Total Revenue, and Customer Satisfaction Score. Active Bikes (8) is the only operational field with a value.

CHECK YOURSELF

If you calculate the average Total Hires across all 42 records and HC-042 has a blank in that column, what effect does the blank have on the result?

Answer: Excel's AVERAGE function ignores blank cells — it calculates the average over 41 values, not 42. This makes the average slightly higher than it would be if the blank were included as zero. Whether this is the right treatment depends on whether Meadow Park genuinely had zero hires or simply did not report.

Spot the Mistake

Marcus has validated the data. Everything is fine. Apparently.

Marcus has sent you this message before the team meeting:

"Just letting you know — I checked the GreenWheels dataset. I sorted it by Record ID (A to Z) and scrolled all the way through. Everything looks perfectly fine to me. No issues at all. You can go straight to the analysis."

"P.S. Do you want me to make a 3D pie chart of the zones? I already made one. It has shadows."

— Marcus

Marcus used the wrong technique. Sorting by Record ID (HC-001, HC-002...) puts the records in the order they were entered. It tells you nothing about whether the numeric values are valid.

TRY IT

Sort the Total Hires column from largest to smallest. What appears at the top? Then sort from smallest to largest. What appears at the top? Check both extreme values against the Data Dictionary.

What should Marcus have done differently?

ANSWER

Sorting by Record ID is alphabetical housekeeping, not validation. Marcus needed to sort by numeric columns — Total Hires, Total Revenue, Avg Duration — to bring extreme values into view.

He also needed to filter for zeros and check for zone inconsistencies. A visual scroll through 40 rows sorted by ID will not show any of the six anomalies in this dataset.

The correct answer for Brian's calculator: there are 6 anomalies, not 0.

Mini-Mission

A two-minute end-of-pack challenge

Priya has sent a message before the analysis begins:

"Quick validation check before we proceed. Three things: how many anomalies did you find, which one would most distort the total hire figures if left unfixed, and what needs to happen before the analysis can start?"

— Priya

You have two minutes. Use the anomaly log you built in Chunk 5.

TRY IT

Write your answer as three short sentences — one per question.

ANSWER

1. Eight anomalies were found across seven records: six data quality issues, one duplicate, and one record with four blank numeric fields.
2. HC-015 (Market Street, Q3 2024) with 1,847 Total Hires would most distort the figures — it is more than double the maximum valid value and would inflate any total or average hire calculation.
3. Each anomaly must be documented in the Anomaly_Log sheet and investigated or flagged before totals and averages are calculated. Pack 3 covers how to use Power Query to clean the data systematically.

You Can Now Answer

By the end of this pack, you should be able to answer each of these questions without looking anything up.

1. How do I sort a column to bring the largest values to the top?
2. How do I use AutoFilter to show only rows where a value is zero?
3. How do I check whether a value is within the valid range stated in the Data Dictionary?
4. How do I spot when two values in the same row contradict each other?
5. How do I check that all records for one station have the same Zone value?
6. What should I write in my anomaly log for each issue I find?
7. Why do I document anomalies rather than just correcting them straight away?
8. How do I find duplicate records where two rows cover the same station and quarter?
9. How do I find blank cells in a column and understand what effect they have on calculations?

If You Got Stuck

Four common stuck-points, and where to go to get unstuck.

"My sort put numbers in the wrong order — 1, 10, 11, 2..." The column is probably formatted as text, not numbers. Go back to Chunk 1. Text sorts differently from numbers — 10 comes before 2 alphabetically.

"My filter did not find any zeros even though I can see one." The cell may contain a space or a text value that displays as zero. Go back to Chunk 2. Try using Number Filters → Equals → 0 rather than filtering for a blank.

"I cannot find the filter arrows." Go back to Chunk 2. Click anywhere inside the data, then press Ctrl+Shift+L to toggle AutoFilter on. The arrows appear in the header row.

"I sorted by Station Name but I cannot see the zone problem." Go back to Chunk 4. After sorting, look at the Zone column next to each group of station rows. South Station's four rows should all say 'South' — one says 'West'.